EXTENSIVE MATHEMATICAL DERIVATIONS

Brownian Bridge Diffusion Model

Accelerated Sampling and Mathematical Consistency

Step-by-step proofs, probability identities,
Gaussian calculations, and mathematical interpretation

PART 4 — EXTENSIONS, CHECKS, AND SUMMARY

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1 Purpose of this report

This report collects the additional mathematical checks and extensions discussed alongside the main BBDM derivations: nonconsecutive-time transitions for accelerated sampling, the Brownian Bridge covariance structure, endpoint conditioning, and common notation pitfalls. It also records which statements follow from the equations and which require the Brownian Bridge Markov construction.

2 Transition between nonconsecutive timesteps

Let $s < t$. The forward marginal at time τ is

$$q(x_\tau \mid x_0, y) = \mathcal{N}(x_\tau;\, a_\tau x_0 + m_\tau y, \delta_\tau I)\,. \tag{1.1}$$

For a Brownian Bridge Markov process, the centered states satisfy

$$\mathbb{E}[B_t \mid B_s] = r_{t|s} B_s, \qquad r_{t|s} = \frac{a_t}{a_s} = \frac{1 - m_t}{1 - m_s}. \tag{1.2}$$

The conditional variance follows from the Gaussian conditioning identity:

$$\begin{aligned} \delta_{t|s} &= \delta_t - r_{t|s}^2 \delta_s \\ &= \delta_t - \delta_s \left(\frac{1 - m_t}{1 - m_s}\right)^2. \end{aligned} \tag{1.3}$$

The corresponding transition mean is

$$r_{t|s} x_s + (m_t - r_{t|s} m_s) y.$$

Hence

$$\boxed{q(x_t \mid x_s, y) = \mathcal{N}\Big(x_t;\, r_{t|s} x_s + (m_t - r_{t|s} m_s) y,\, \delta_{t|s} I\Big)\,.} \tag{1.4}$$

This is the same transition structure as the consecutive-step case, with $t - 1$ replaced by the earlier selected timestep s.

2.1 Posterior for a skipped step

Bayes' theorem gives

$$q(x_s \mid x_t, x_0, y) \propto q(x_t \mid x_s, y) q(x_s \mid x_0, y). \tag{1.5}$$

This is again a product of two Gaussian densities. Repeating the precision-addition calculation from Part 2, define

$$r = \frac{1 - m_t}{1 - m_s}, \quad b = m_t - r m_s, \quad v = \delta_t - r^2 \delta_s,$$

and

$$\mu_s = a_s x_0 + m_s y.$$

Then

$$\boxed{q(x_s \mid x_t, x_0, y) = \mathcal{N}(x_s; \widetilde{\mu}_{t,s}, \widetilde{\delta}_{t,s} I),} \tag{1.6}$$

where

$$\boxed{\widetilde{\delta}_{t,s} = \frac{v\delta_s}{\delta_t}} \tag{1.7}$$

and

$$\boxed{\widetilde{\mu}_{t,s} = \frac{\delta_s r}{\delta_t}(x_t - by) + \frac{v}{\delta_t}(a_s x_0 + m_s y).} \tag{1.8}$$

The derivation is identical to the consecutive-step posterior: the likelihood has variance vI, the prior has variance $\delta_s I$, and their precisions add.

3 Brownian Bridge construction and covariance

A standard one-dimensional Brownian Bridge from endpoint x_0 to endpoint y over normalized time $u \in [0, 1]$ can be represented as

$$X(u) = (1 - u)x_0 + uy + W(u) - uW(1), \tag{2.1}$$

where $W(u)$ is standard Brownian motion. The stochastic bridge component

$$B(u) = W(u) - uW(1)$$

is zero at both endpoints. Its covariance is

$$\begin{aligned} \mathrm{Cov}(B(u), B(v)) &= \mathrm{Cov}(W(u) - uW(1), W(v) - vW(1)) \\ &= \min(u, v) - vu - uv + uv \\ &= \min(u, v) - uv. \end{aligned} \tag{2.2}$$

For $u \le v$, this becomes

$$\mathrm{Cov}(B(u), B(v)) = u(1 - v). \tag{2.3}$$

Its marginal variance is

$$\mathrm{Var}(B(u)) = u(1 - u). \tag{2.4}$$

For $u \le v$, dividing the cross-covariance by the earlier-time variance gives

$$\frac{\mathrm{Cov}(B(v), B(u))}{\mathrm{Var}(B(u))} = \frac{u(1 - v)}{u(1 - u)} = \frac{1 - v}{1 - u}. \tag{2.5}$$

For jointly Gaussian variables, the conditional mean is this regression coefficient times the conditioning variable:

$$\mathbb{E}[B(v) \mid B(u)] = \frac{1 - v}{1 - u} B(u). \tag{2.6}$$

The conditional variance is

$$\begin{aligned}\mathrm{Var}(B(v) \mid B(u)) &= v(1-v) - \frac{u^2(1-v)^2}{u(1-u)} \\ &= v(1-v) - \frac{u(1-v)^2}{1-u} \\ &= \frac{(v-u)(1-v)}{1-u}.\end{aligned} \tag{2.7}$$

This establishes the standard Brownian Bridge transition and explains the contraction ratio used in the discrete bridge construction.

4 How the BBDM schedule fits the bridge

BBDM uses the interpolation schedule

$$m_t = \frac{t}{T}$$

and variance schedule

$$\delta_t = 2s\, m_t(1-m_t). \tag{3.1}$$

The standard Brownian Bridge has variance $u(1-u)$; the BBDM schedule scales this variance by $2s$. The bridge contraction factor for $s < t$ is

$$r_{t|s} = \frac{1-m_t}{1-m_s}. \tag{3.2}$$

Consequently,

$$\delta_{t|s} = \delta_t - r_{t|s}^2 \delta_s. \tag{3.3}$$

For $m_t = t/T$ and $\delta_t = 2s\, m_t(1-m_t)$, this yields

$$\delta_{t|s} = 2s\, \frac{(m_t - m_s)(1-m_t)}{1-m_s}. \tag{3.4}$$

For consecutive steps $s = t-1$,

$$\delta_{t|t-1} = \frac{2s(T-t)}{T(T-t+1)}. \tag{3.5}$$

The scalar s in the variance schedule should not be confused with the earlier timestep label often also denoted s; when both occur in one derivation, rename the variance-scale hyperparameter, for example λ, to avoid ambiguity.

5 The ratio r_t: exactly what it means

The deterministic bridge interpolation gives

$$x_t - y = (1-m_t)(x_0 - y) + B_t.$$

The deterministic displacement from the endpoint at time t is

$$a_t(x_0 - y),$$

and at time $t-1$ it is

$$a_{t-1}(x_0 - y).$$

Their ratio is

$$\frac{a_t(x_0 - y)}{a_{t-1}(x_0 - y)} = \frac{a_t}{a_{t-1}} = \boxed{\frac{1 - m_t}{1 - m_{t-1}}}. \tag{4.1}$$

This motivates the notation r_t. But one must be precise: this deterministic ratio alone does not prove the stochastic conditional expectation $\mathbb{E}[B_t \mid B_{t-1}] = r_t B_{t-1}$. That additional statement follows from the Brownian Bridge covariance/Markov structure, as derived in Part 1 and in Section 2 above.

6 Conditioning notation: why x_0 disappears from one factor

A fully explicit Bayes expression is

$$\boxed{q_{BB}(x_{t-1} \mid x_t, x_0, y) = \frac{q_{BB}(x_t \mid x_{t-1}, x_0, y) q_{BB}(x_{t-1} \mid x_0, y)}{q_{BB}(x_t \mid x_0, y)}}. \tag{5.1}$$

By the Markov property,

$$q_{BB}(x_t \mid x_{t-1}, x_0, y) = q_{BB}(x_t \mid x_{t-1}, y). \tag{5.2}$$

Thus the abbreviated expression

$$q_{BB}(x_{t-1} \mid x_t, x_0, y) \propto q_{BB}(x_t \mid x_{t-1}, y) q_{BB}(x_{t-1} \mid x_0, y) \tag{5.3}$$

is valid. The first factor omits x_0 because the one-step Markov transition depends on the present state and the fixed target endpoint, not on the earlier source endpoint once the present state is given. The second factor still explicitly depends on x_0.

7 Mathematical consistency checklist

When checking a derivation or a printed equation, verify the following points.

1. **Variance versus standard deviation.** If the marginal covariance is $\delta_t I$, the sample equation contains $\sqrt{\delta_t}\epsilon_t$, not $\delta_t \epsilon_t$.
2. **Marginal noise versus innovation.** The ϵ_t variables at different times are generally correlated in a bridge. The fresh transition innovation z_t is independent of the previous state.
3. **Transition variance.** The correct conditional variance is $\delta_t - r_t^2 \delta_{t-1}$, not $\delta_t + r_t^2 \delta_{t-1}$. The subtraction is the Gaussian conditional-variance formula.

4. **Markov notation.** x_0 may be suppressed in $q(x_t \mid x_{t-1}, x_0, y)$ only after invoking the Markov property.
5. **Posterior conditioning.** The posterior still conditions on x_0, because its prior factor $q(x_{t-1} \mid x_0, y)$ depends on the source endpoint.
6. **ELBO terms.** The terminal KL is constant with respect to θ only if both the forward process and terminal prior are fixed.
7. **Noise target.** Standardized noise ϵ and the residual $x_t - x_0$ are not the same quantity. A model's coefficient and sign must match its chosen target.
8. **Nonconsecutive steps.** Accelerated sampling uses the same bridge conditioning formulas with arbitrary selected times $s < t$; it is not a new forward process.

8 Consolidated equation sheet

$$x_t = (1 - m_t)x_0 + m_t y + \sqrt{\delta_t}\epsilon_t, \tag{1}$$

$$r_t = \frac{1 - m_t}{1 - m_{t-1}}, \tag{2}$$

$$b_t = m_t - r_t m_{t-1}, \tag{3}$$

$$\delta_{t|t-1} = \delta_t - r_t^2 \delta_{t-1}, \tag{4}$$

$$q(x_t \mid x_{t-1}, y) = \mathcal{N}(x_t; r_t x_{t-1} + b_t y, \delta_{t|t-1} I), \tag{5}$$

$$\widetilde{\delta}_t = \frac{\delta_{t|t-1}\delta_{t-1}}{\delta_t}, \tag{6}$$

$$\widetilde{\mu}_t = \frac{\delta_{t-1} r_t}{\delta_t}(x_t - b_t y) + \frac{\delta_{t|t-1}}{\delta_t}\big((1 - m_{t-1})x_0 + m_{t-1} y\big), \tag{7}$$

$$\delta_{t|s} = \delta_t - \delta_s \left(\frac{1 - m_t}{1 - m_s}\right)^2. \tag{8}$$

9 Final conceptual picture

The Brownian Bridge is not simply a linear interpolation with independent noise added independently at every timestep. Its stochastic component is a correlated Gaussian process constrained by two endpoints. The Markov transition is obtained from the joint covariance structure, and its conditional variance contains a subtraction term. Bayes' theorem then combines this transition likelihood with the previous-time marginal to produce a Gaussian posterior. Finally, the ELBO trains a reverse model to match those posterior transitions, and equal-covariance Gaussian KL terms reduce to weighted squared errors. Accelerated sampling follows by applying the same conditional-Gaussian calculations to nonconsecutive timesteps.